8. The decimal representation of 30 *kg* and 43 *g* is _____ *kg*.

(i) 30.43 (ii) 30.430 (iii) 30.043 (iv) 30.0043

9. A cricket pitch is about 264 cm wide. It is equal to _____ *m*.

(i) 26.4 (ii) 2.64 (iii) 0.264 (iv) 0.0264

1.3 Fractions and Decimals

Let us see the relationship between fractions and decimals.

1.3.1 Conversion of Fractions to Decimals

We are familiar with fraction as a part of a whole. The place value of the decimal digits of a number are tenths$\left(\frac{1}{10}\right)$, hundredths$\left(\frac{1}{100}\right)$, thousandths$\left(\frac{1}{1000}\right)$ and so on.

If the denominator of a fraction is any of $10, 10^2, 10^3, \ldots$ we can express them as decimals. Consider the example of distributing a box of 10 pencils to ten students. The portion of pencils given to 6 students will be $\frac{6}{10}$ which can be expressed as 0.6.

If the denominator of a fraction is any number that can be made as powers of 10 using the concept of equivalent fractions, then it can also be expressed as decimals. Consider the example of sharing 5 peanut cakes among five friends. The share of one person is $\frac{1}{5}$. To represent this fraction as a decimal number, we first convert the denominator into 10. This can be done by writing the equivalent fraction of $\frac{1}{5}$, namely $\frac{2}{10}$. Now the decimal representation of $\frac{2}{10}$ is 0.2

Think

Can you express the denominators of all fractions as powers of 10?

1.3.2 Conversion of Decimals to Fractions

As we convert fractions into a decimal number, the decimal numbers can also be expressed as fractions.

For example, let the price of brand '*x*' slippers be ₹ 399.95

Expanding the above price, we get,

$$399.95 = 3\times100 + 9\times10 + 9\times1 + 9\times\frac{1}{10} + 5\times\frac{1}{100}$$

$$= 399 + \frac{95}{100} = \frac{39995}{100} = \frac{7999}{20}$$

Similarly, if the price of brand '*y*' slipper is ₹ 159.95, then it can be expressed in terms of fractions as below.

$$159.95 = 159 + \frac{95}{100} = \frac{15995}{100} = \frac{3199}{20}$$

7th_Maths_T2_EM_Chp1.indd 9 09-07-2019 12:48:40